A PROOFS OF THEOREMS FROM § 3

A.1 Lemma 3.9

For any relation $R \subseteq X \times Y$ the functions $\triangle_R$ and $\triangledown_R$ are conjugate.

Proof. Let $X' \in \mathbb{P}(X)$, $Y' \in \mathbb{P}(Y)$.

WTS: $\triangle_R(X') \cap Y' = \varnothing \implies X' \cap \triangledown_R(Y') = \varnothing$

$\triangle_R(X') \cap Y' = \varnothing$	suppose
$\{y \in Y \mid \exists x \in X'.(x,y) \in R\} \cap Y' = \varnothing$	def of $\triangle_R$
$\{y \in Y' \mid \exists x \in X'.(x,y) \in R\} = \varnothing$	def of $\cap$ (1)
$\forall y \in Y'.\ \forall x \in X'.(x,y) \notin R$	equivalent to (1) (2)
$\{x \in X' \mid \exists y \in Y'.(x,y) \in R\} = \varnothing$	equivalent to (2)
$\leftrightarrows X' \cap \{x \in X \mid \exists y \in Y'.(x,y) \in R\} = \varnothing$	def of $\cap$
$X' \cap \triangledown_R(Y') = \varnothing$	def of $\triangledown_R$

WTS: $X' \cap \triangledown_R(Y') = \varnothing \implies \triangle_R(X') \cap Y' = \varnothing$

$X' \cap \triangledown_R(Y') = \varnothing$	suppose
$X' \cap \{x \in X \mid \exists y \in Y'.(x,y) \in R\} = \varnothing$	def of $\triangledown_R$
$\{x \in X' \mid \exists y \in Y'.(x,y) \in R\}$	def of $\cap$ (3)
$\forall y \in Y'.\ \forall x \in X'.(x,y) \notin R$	equivalent to (3) (4)
$\leftrightarrows \{y \in Y \mid \exists x \in X'.(x,y) \in R\} \cap Y' = \varnothing$	equivalent to (4)
$\triangle_R(X') \cap Y' = \varnothing$	def of $\triangle_R$

□

A.2 Proposition 3.15

Suppose functions $f : A \to B$ and $g : B \to A$ between Boolean algebras. The following statements are equivalent:

(1) f and g form a conjugate pair
(2) f and g° form a Galois connection

Proof. Fix $x \in A$ and $y \in B$.

WTS: f, g conjugate $\implies$ f, g° form Galois connection.

$f(x) \leq_B y$	(suppose)
$f(x) \wedge_B \neg y = \bot$	
$x \wedge_A g(\neg y) = \bot$	(conjugacy)
$x \leq_A \neg(g(\neg y))$	
$\leftrightarrows x \leq_A g^\circ(y)$	(def of $\cdot^\circ$)

WTS: f, g° Galois connection $\implies$ f, g conjugate

$f(x) \wedge_B y = \bot$	suppose
$f(x) \leq_B \neg y$	
$x \leq_A g^\circ(\neg y)$	(GC)

$x \leq_A \neg(g(y))$ (def of $\cdot^\circ$)

$x \wedge_A g(y) = \bot$

□

A.3 Lemma 3.13

Fix $R \subseteq X \times Y$. We proceed by direct calculation.

Proof. **WTS:** $\triangle_R^\circ = \blacktriangle_R$:

$\triangle_R(X')^\circ = \{y \in Y \mid \exists x \in X'.(x,y) \in R\}^\circ$ def of $\triangle_R$

$= \{y \in Y \mid \exists x \in (X')^c.(x,y) \in R\}^c$ def of $\cdot^\circ$

$= \{y \in Y \mid \exists x \in (X \setminus X').(x,y) \in R\}^c$ def of $\cdot^c$

$= \{y \in Y \mid \nexists x \in (X \setminus X').(x,y) \in R\}$ $\cdot^c$ negates predicate

$= \blacktriangle_R(X')$ def of $\blacktriangle_R$

WTS: $\triangledown_R^\circ = \blacktriangledown_R$:

$\triangledown_R(Y')^\circ = \{x \in X \mid \exists y \in Y'.(x,y) \in R\}^\circ$ def of $\triangledown_R^\circ$

$= \{x \in X \mid \exists y \in (Y')^c.(x,y) \in R\}^c$ def of $\cdot^\circ$

$= \{x \in X \mid \exists y \in (Y \setminus Y').(x,y) \in R\}^c$ def of $\cdot^c$

$= \{x \in X \mid \nexists y \in (Y \setminus Y').(x,y) \in R\}$ $\cdot^c$ negates predicate

$= \blacktriangledown_R(Y')$ def of $\blacktriangledown_R$

□

A.4 Proposition 3.17

Proof.

⇒ direction:

$X \subseteq \mathsf{S}(G)$ with X demBy$_G$ Y suppose (1)

$$\frac{(X,\varnothing), G \text{ demByV } H}{X \text{ demBy}_G\ \mathsf{T}(H) = Y}$$ inversion; $\exists H$

➪ $\triangle_G(X) = \mathsf{T}(G) \cap \triangle_G^*(X) = Y$ (1); A.2; Proposition A.5

⇐ direction:

$X \subseteq \mathsf{S}(G)$ with $\triangle_G(X) = \mathsf{T}(G) \cap \triangle_G^*(X) = Y$ suppose

$(X,\varnothing), G$ demByV H with $\mathsf{T}(H) = Y$ Proposition A.5; $\exists H$

➪ $$\frac{(X,\varnothing), G \text{ demByV } H}{X \text{ demBy}_G\ \mathsf{T}(H) = Y}$$ def. demBy

□

A.5 Correctness of demByV

The *output* relation of a graph G with reachability relation R, defined as $R \cap \mathsf{V}(G) \times \mathsf{T}(G)$, gives rise to the following extension of $\triangle_G$:

Definition A.1 ($\triangle^*$ for a dependence graph). For a graph G with output relation R, define:

$$\triangle^*_G := \triangle_R : \mathbb{P}(\mathsf{V}(G)) \to \mathbb{P}(\mathsf{V}(G))$$

Lemma A.2. $\triangle_G = \mathsf{T}(G) \cap \triangle^*_G(X)$.

Lemma A.3. *If* $X \subseteq \mathsf{T}(G)$ *then* $\triangle^*_G(X) = X$.

Lemma A.4. *Suppose* $\mathsf{V}(H) \subseteq \mathsf{V}(G)$. *For any* $\alpha \in \mathsf{T}(H)$ *and any* $E = \mathsf{outE}_G(\alpha) \neq \emptyset$:

$$\triangle^*_G(\mathsf{T}(H)) = \triangle^*_{G \setminus_\mathsf{E} E}(\mathsf{T}(H \cup E))$$

Proposition A.5. *For* $\mathsf{V}(H) \subseteq \mathsf{V}(G)$, *and any* $Y \subseteq \mathsf{T}(G)$:

$$(\exists H'.\ H, G\ \mathsf{demByV}\ H'\ \textit{with}\ \mathsf{T}(H') = Y) \iff \mathsf{T}(H \cup G) \cap \triangle^*_G(\mathsf{T}(H)) = Y$$

Proof.

$\mathsf{V}(H) \subseteq \mathsf{V}(G)$ and $Y \subseteq \mathsf{T}(G)$ — suppose

$\Rightarrow$ direction:

$H, G\ \mathsf{demByV}\ H'$ with $\mathsf{T}(H') = Y$ — suppose; $\exists H'$

By induction on derivation of demByV.

Subcase $\dfrac{\text{done}}{H, G\ \mathsf{demByV}\ H = H'}\ \mathsf{T}(H) \subseteq \mathsf{T}(G)$

$\Rightarrow \mathsf{T}(H \cup G) \cap \triangle^*_G(\mathsf{T}(H)) = \mathsf{T}(H) \cap \triangle^*_G(\mathsf{T}(H)) = \mathsf{T}(H)$ — Lemma A.3

Subcase $\dfrac{\text{extend}\quad H \cup E, G \setminus_\mathsf{E} E\ \mathsf{demByV}\ H'}{H, G\ \mathsf{demByV}\ H'}\ \alpha \in \mathsf{T}(H) \wedge E = \mathsf{outE}_G(\alpha) \neq \emptyset$

$\triangle^*_{G \setminus_\mathsf{E} E}(\mathsf{T}(H \cup E)) = \mathsf{T}(H')$ — IH

$\Rightarrow \triangle^*_G(\mathsf{T}(H)) = \mathsf{T}(H')$ — Lemma A.4

$\Leftarrow$ direction:

By strong induction over $|\mathsf{E}(G)|$.

$\mathsf{T}(G \cup H) \cap \triangle^*_G(\mathsf{T}(H)) = Y$ — suppose (2)

Subcase $\mathsf{T}(H) \subseteq \mathsf{T}(G)$.

$\dfrac{\text{done}}{H, G\ \mathsf{demByV}\ H}\ \mathsf{T}(H) \subseteq \mathsf{T}(G)$ — def. demByV

$\Rightarrow \mathsf{T}(G \cup H) \cap \mathsf{T}(H) = \mathsf{T}(G \cup H) \cap \triangle^*_G(\mathsf{T}(H)) = Y$ — Lemma A.3; (2)

Subcase $\alpha \in \mathsf{T}(H) \wedge E = \mathsf{outE}_G(\alpha) \neq \emptyset$. — $\exists \alpha, \exists E$

$\mathsf{E}(G \setminus_\mathsf{E} E) \subset \mathsf{E}(G)$

$H \cup E, G \setminus_\mathsf{E} E\ \mathsf{demByV}\ H'$ with $\mathsf{T}(H') = \mathsf{T}((G \setminus_\mathsf{E} E) \cup (H \cup E)) \cap \triangle^*_{G \setminus_\mathsf{E} E}(\mathsf{T}(H \cup E))$ — IH; $\exists H'$

$\mathsf{T}(H') = \mathsf{T}(G \cup H) \cap \triangle^*_G(\mathsf{T}(H)) = Y$ — Lemma A.4

$\Rightarrow$ $\dfrac{\text{extend}\quad H \cup E, G \setminus_\mathsf{E} E\ \mathsf{demByV}\ H'}{H, G\ \mathsf{demByV}\ H'}\ \alpha \in \mathsf{T}(H) \wedge E = \mathsf{outE}_G(\alpha) \neq \emptyset$ — def. demByV

□

A.6 Proposition 3.19

Proof.

⇒ direction:

$X \subseteq \mathsf{S}(G)$ and $Y \subseteq \mathsf{T}(G)$ with X suff_G Y — suppose

$$\frac{(X,\varnothing),(\varnothing,\varnothing),G \ \mathsf{suffE}\ H}{X \ \mathsf{suff}_G \ \mathsf{V}(H) \cap \mathsf{T}(G) = Y}$$ — inversion; $\exists H$

$(X,\varnothing),(\varnothing,\varnothing),G$ form a partial slice of G — immediate

➪ $\blacktriangle_G(X) = \mathsf{T}(G) \cap \blacktriangle_G^*(X) = Y$ — Lemma A.10, Proposition A.14

⇐ direction:

$X \subseteq \mathsf{S}(G)$ and $Y \subseteq \mathsf{T}(G)$ with $\blacktriangle_G(X) = Y$ — suppose

$\blacktriangle_G(X) = \mathsf{T}(G) \cap \blacktriangle_G^*(X)$ — Lemma A.10

$(X,\varnothing),(\varnothing,\varnothing),G$ form a partial slice of G — immediate

$(X,\varnothing),(\varnothing,\varnothing),G \ \mathsf{suffE}\ H$ — Proposition A.14; $\exists H$

➪ $$\frac{(X,\varnothing),(\varnothing,\varnothing),G \ \mathsf{suffE}\ H}{X \ \mathsf{suff}_G \ \mathsf{V}(H) \cap \mathsf{T}(G) = Y}$$ — def. suff

□

A.7 Correctness of suffE

Definition A.6 (Partial Slice). Graphs H, P, G constitute a *partial slice* of the graph G_0 iff H, P, G form a partition of G_0 and:

(i) $\mathsf{V}(H) \subseteq \mathsf{V}(P) \subseteq \mathsf{V}(G) = \mathsf{V}(G_0)$
(ii) $\mathsf{V}(H) \supseteq \mathsf{T}(P)^c$
(iii) $\mathsf{V}(H) = \mathsf{S}(P)$
(iv) $\mathsf{V}(P) = \mathsf{S}(P) \cap \mathsf{T}(P)$
(v) $\forall(\alpha,\beta) \in (\mathsf{S}(G_0) \setminus \mathsf{S}(H)) \times \mathsf{V}(H).\ (\alpha,\beta) \notin R_{G_0}$
(vi) $\mathsf{S}(G) \setminus (\mathsf{T}(P) \setminus \mathsf{S}(P)) = \mathsf{V}(H) \uplus (\mathsf{S}(G_0) \setminus \mathsf{S}(H))$

Corollary A.7. *H, P, G a partial slice $\iff$ H, P, G edge disjoint and* $\forall(\alpha,\beta) \in \mathsf{E}(G) \cup \mathsf{E}(P).\ \beta \notin \mathsf{V}(H)$

Definition A.8 (Input Relation). For a graph G with reachability relation R, define it's input relation $R_I := (\mathsf{S}(G) \times \mathsf{V}(G)) \cap R$

Definition A.9 ($\blacktriangle^*$ for a dependence graph). For a graph G, with an input relation R_I, and a set of input vertices $X \subseteq \mathsf{S}(G)$:

$$\blacktriangle_G^*(X) := \blacktriangle_{R_I}(X) = \{y \in \mathsf{V}(G) \mid \nexists x \in \mathsf{S}(G) \setminus X.\ (x,y) \in R_I\}$$

Lemma A.10. $\blacktriangle_G(X) = \mathsf{T}(G) \cap \blacktriangle_G^*(X)$

Lemma A.11. *If H, P, G a partial slice of G_0 with $\mathsf{S}(G) \cap \mathsf{V}(P) \subseteq \mathsf{S}(P)$ and $\mathsf{V}(H) \subseteq \mathsf{T}(G)$:*

$$\blacktriangle_{G_0}^*(\mathsf{S}(H)) = \mathsf{V}(H)$$

Proof.

H, P, G a partial slice — suppose

$\mathsf{S}(G) \cap \mathsf{V}(P) \subseteq \mathsf{S}(P)$ — suppose (1)

$\mathsf{V}(H) \subseteq \mathsf{T}(G)$ — suppose (2)

$\subseteq$ **direction**

$y \in \blacktriangle^*_{G_0}(\mathsf{S}(H))$ suppose

$\forall x \in \mathsf{S}(G_0) \setminus \mathsf{S}(H).\ (x, y) \notin R_{G_0}$ def. $\blacktriangle^*$

$y \in \mathsf{V}(G)$

$x \in \mathsf{S}(G).\ (x, y) \in R_G$ $\exists x$, G acyclic

$x \in \mathsf{V}(H) \cup \mathsf{S}(G_0) \setminus \mathsf{S}(H) \cup \mathsf{T}(P) \setminus \mathsf{S}(P)$ H, P, G a partial slice

$x \notin \mathsf{S}(G_0) \setminus \mathsf{S}(H)$ immediate

$x \notin \mathsf{T}(P) \setminus \mathsf{S}(P)$ contradicts 1

$\Rightarrow x \in \mathsf{V}(H)$ $\mathsf{S}(G)$ invariant

$x \neq y \implies x \notin \mathsf{T}(H) \implies \bot$ contradicts 2

$\Rightarrow x = y$

$\Rightarrow y \in \mathsf{V}(H)$

$\supseteq$ **direction**

$y \in \mathsf{V}(H)$ suppose

$y \in \mathsf{V}(G_0)$

$\mathsf{S}(G_0) \setminus \mathsf{S}(H) \subseteq (\mathsf{V}(G \cup P) \setminus \mathsf{V}(H))$

$\forall \alpha \in \mathsf{S}(G_0) \setminus \mathsf{S}(H).\ (\alpha, y) \notin R_{G_0}$ def. partial slice

$\Rightarrow y \in \blacktriangle^*_{G_0}(\mathsf{S}(H))$

□

Lemma A.12.

(1) If $\alpha \in \mathsf{V}(H)$, $E = \mathsf{outE}_G(\alpha) \neq \varnothing$ then:
H, P, G a partial slice of G_0 $\iff$ $H, P \cup E, G \setminus_{\mathsf{E}} E$ a partial slice of G_0

(2) If $\alpha \in \mathsf{S}(G)$, $E = \mathsf{inE}_P(\alpha) \neq \varnothing$ then: H, P, G a partial slice G_0 $\iff$ $H \cup E, P \setminus_{\mathsf{E}} E, G$ a partial slice of G_0

Proof.

(1.II), $\Rightarrow$

$E = \{\alpha\} \times B$ suppose

$\mathsf{T}(P)^c \subseteq \mathsf{V}(H)$ suppose

$\mathsf{T}(P \cup E)^c = \mathsf{T}(P)^c \cup \{\alpha\}$ def. E

$\Rightarrow \mathsf{T}(P \cup E)^c \subseteq \mathsf{V}(H)$ $\alpha \in \mathsf{V}(H)$

(1.II), $\Leftarrow$

$\mathsf{T}(P \cup E)^c \subseteq \mathsf{V}(H)$ suppose

$\mathsf{T}(P)^c \subseteq \mathsf{T}(P \cup E)^c$

$\Rightarrow \mathsf{T}(P)^c \subseteq \mathsf{V}(H)$

(1.III),$\Rightarrow$

$E = \{\alpha\} \times B$ suppose

$\mathsf{V}(H) = \mathsf{S}(P)$ suppose

Need to show: $\mathsf{V}(H) = \mathsf{S}(P \cup E)$

$\mathsf{S}(P \cup E) = \mathsf{S}(P) \setminus (\mathsf{S}(P) \cap B)$

Need to show: $\mathsf{S}(P) \cap B = \varnothing$

$(\alpha, \beta) \in \mathsf{E}(G)$ $\forall (\alpha, \beta) \in E$

$\beta \in \mathsf{S}(P) = \mathsf{V}(H) \implies \bot$ Contra partial slice(V)

$\leadsto \beta \notin \mathsf{S}(G)$

$\leadsto \mathsf{S}(P) \cap \beta = \varnothing$

$\leadsto \mathsf{V}(H) = \mathsf{S}(P \cup E)$

(1.III), $\Leftarrow$

$\mathsf{V}(H) = \mathsf{S}(P \cup E)$ suppose

$E = \{\alpha\} \times B$ suppose

Want to show: $\mathsf{V}(H) = \mathsf{S}(P)$

$\mathsf{S}(P) = \mathsf{S}(P \cup E) \setminus B$

suffices to show: $\mathsf{S}(P \cup E) \cap B = \varnothing$

$(\alpha, \beta) \in E$. $(\alpha, \beta) \in \mathsf{E}(G)$

$\leadsto \beta \notin \mathsf{V}(H) = \mathsf{S}(P \cup E)$ partial slice (V)

$\leadsto \mathsf{S}(P \cup E) \cap B = \varnothing$

$\leadsto \mathsf{S}(P) = \mathsf{S}(P \cup E)$

$\leadsto \mathsf{V}(H) = \mathsf{S}(P)$

(1.IV), $\Rightarrow$

$\mathsf{V}(P) = \mathsf{S}(P) \cup \mathsf{T}(P)$ suppose

$E = B \times \{\alpha\}$ suppose

$\mathsf{V}(P \cup E) = \mathsf{V}(P) \cup B$

suffices to show that:$\mathsf{S}(P) \cup \mathsf{T}(P) \cup B = \mathsf{S}(P \cup E) \cup \mathsf{T}(P \cup E)$

$\mathsf{S}(P) = \mathsf{S}(P \cup E)$ proved in proof of (iii)

$\alpha \in \mathsf{S}(P) = \mathsf{S}(P \cup E)$ $\alpha \in \mathsf{V}(H)$

$\mathsf{T}(P \cup E) = (\mathsf{T}(P) \cup B) \setminus \alpha$ def. E

$\leadsto \mathsf{S}(P) \cup \mathsf{T}(P) \cup B = \mathsf{S}(P \cup E) \cup ((\mathsf{T}(P) \cup B) \setminus \{\alpha\})$

$\leadsto \mathsf{S}(P) \cup \mathsf{T}(P) \cup B = \mathsf{S}(P \cup E) \cup \mathsf{T}(P \cup E)$

(1.IV), $\Leftarrow$

$\mathsf{V}(P \cup E) = \mathsf{S}(P \cup E) \cup \mathsf{T}(P \cup E)$ suppose

$\mathsf{V}(P \cup E) = \mathsf{V}(P) \cup B$

$\mathsf{S}(P \cup E) \cup \mathsf{T}(P \cup E) = \mathsf{V}(P) \cup B$

$\mathsf{S}(P \cup E) = \mathsf{S}(P)$ def. E

$\mathsf{T}(P \cup E) = (\mathsf{T}(P) \cup B) \setminus \{\alpha\}$ def. E

$B \cap \mathsf{S}(P) = \varnothing$

$\mathsf{S}(P) \setminus B = \mathsf{S}(P)$

$(\mathsf{S}(P \cup E) \cup \mathsf{T}(P \cup E)) \setminus B = \mathsf{V}(P)$

$(\mathsf{S}(P \cup E) \setminus B) \cup ((\mathsf{T}(P) \cup B \setminus \{\alpha\}) \setminus B) = \mathsf{V}(P)$

$\leadsto \mathsf{S}(P) \cup (\mathsf{T}(P) \setminus \{\alpha\}) = \mathsf{V}(P)$

$\leadsto \mathsf{S}(P) \cup \mathsf{T}(P) = \mathsf{V}(P)$ $\alpha \in \mathsf{S}(P)$

(1.V), $\Rightarrow$

H, P, G a partial slice suppose

$\forall (\alpha, \beta) \in \mathsf{E}(G) \cup \mathsf{E}(P)$. $\beta \notin \mathsf{V}(H)$ def. partial slice

$\mathsf{E}(G) \cup \mathsf{E}(P) = \mathsf{E}(G \setminus_{\mathsf{E}} E) \cup \mathsf{E}(P \cup E)$

$\Rightarrow \forall(\alpha, \beta) \in \mathsf{E}(G \setminus_{\mathsf{E}} E) \cup \mathsf{E}(P \cup E).\ \beta \notin \mathsf{V}(H)$

(1.V), $\Leftarrow$

$H, P \cup E, G \setminus_{\mathsf{E}} E$ a partial slice — suppose

$\forall(\alpha, \beta) \in \mathsf{E}(G \setminus_{\mathsf{E}} E) \cup \mathsf{E}(P \cup E).\ \beta \notin \mathsf{V}(H)$ — def. partial slice

$\mathsf{E}(G) \cup \mathsf{E}(P) = \mathsf{E}(G \setminus_{\mathsf{E}} E) \cup \mathsf{E}(P \cup E)$

$\Rightarrow \forall(\alpha, \beta) \in \mathsf{E}(G) \cup \mathsf{E}(P).\ \beta \notin \mathsf{V}(H)$

(1.VI)

Suffices to show: $\mathsf{S}(G) \setminus (\mathsf{T}(P) \setminus \mathsf{S}(P)) = \mathsf{S}(G \setminus_{\mathsf{E}} E) \setminus (\mathsf{T}(P \cup E) \setminus \mathsf{S}(P \cup E))$

$\mathsf{S}(G) \setminus (\mathsf{T}(P) \setminus \mathsf{S}(P)) = \mathsf{V}(H) \uplus (\mathsf{S}(G_0) \setminus \mathsf{S}(H))$ — suppose

$E = \{\alpha\} \times B$

Subcase $x \in \mathsf{S}(G) \setminus (\mathsf{T}(P) \setminus \mathsf{S}(P))$.

$x \in \mathsf{S}(G) \subseteq \mathsf{S}(G \setminus_{\mathsf{E}} E)$

$x \notin \mathsf{T}(P) \setminus \mathsf{S}(P)$ (3)

$x \notin B$ — $x \in \mathsf{S}(G)$

$\Rightarrow x \notin \mathsf{T}(P \cup E) \setminus \mathsf{S}(P \cup E)$

$x \in \mathsf{S}(G \setminus_{\mathsf{E}} E) \setminus (\mathsf{T}(P \cup E) \setminus \mathsf{S}(P \cup E))$

$\Rightarrow \mathsf{S}(G) \setminus \mathsf{T}(P) \setminus \mathsf{S}(P) \subseteq \mathsf{S}(G \setminus_{\mathsf{E}} E) \setminus (\mathsf{T}(P \cup E) \setminus \mathsf{S}(P \cup E))$ (4)

Subcase $x \notin \mathsf{S}(G) \setminus (\mathsf{T}(P) \setminus \mathsf{S}(P))$.

Subcase $x \in \mathsf{S}(G)$.

$x \in \mathsf{T}(P) \setminus \mathsf{S}(P)$ (5)

$x \in \mathsf{T}(P \cup E) \setminus \mathsf{S}(P \cup E)$

$\Rightarrow x \notin \mathsf{S}(G \setminus_{\mathsf{E}} E) \setminus (\mathsf{T}(P \cup E) \setminus \mathsf{S}(P \cup E))$

Subcase $x \notin \mathsf{S}(G) \wedge x \in \mathsf{S}(G \setminus_{\mathsf{E}} E)$.

$x \notin \mathsf{S}(G) \setminus (\mathsf{T}(P) \setminus \mathsf{S}(P))$

$(\alpha, x) \in E$ — $x \in \mathsf{S}(G \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(G)$ (6)

$x \in (\mathsf{T}(P \cup E) \setminus \mathsf{S}(P \cup E))$

$\Rightarrow x \notin \mathsf{S}(G \setminus_{\mathsf{E}} E) \setminus (\mathsf{T}(P \cup E) \setminus \mathsf{S}(P \cup E))$

$\Rightarrow \mathsf{S}(G \setminus_{\mathsf{E}} E) \setminus (\mathsf{T}(P \cup E) \setminus \mathsf{S}(P \cup E)) \subseteq \mathsf{S}(G) \setminus (\mathsf{T}(P) \setminus \mathsf{S}(P))$ (7)

$\Rightarrow \mathsf{S}(G) \setminus \mathsf{T}(P) \setminus \mathsf{S}(P) = \mathsf{S}(G \setminus_{\mathsf{E}} E) \setminus (\mathsf{T}(P \cup E) \setminus \mathsf{S}(P \cup E))$

(2.II), $\Rightarrow$

$E = B \times \{\alpha\}$ — suppose

$\mathsf{T}(P)^c \subseteq \mathsf{V}(H)$ — suppose

$\mathsf{V}(H \cup E) = \mathsf{V}(H) \cup \{\alpha\}$

$\mathsf{T}(P \setminus_{\mathsf{E}} E)^c \cup B = \mathsf{T}(P)^c$

$B \subseteq \mathsf{T}(P)^c \subseteq \mathsf{V}(H)$

$\Rightarrow \mathsf{T}(P \setminus_{\mathsf{E}} E)^c \subseteq \mathsf{V}(H) \cup \{\alpha\}$

(2.II), $\Leftarrow$

$\mathsf{T}(P \setminus_{\mathsf{E}} E)^c \subseteq \mathsf{V}(H \cup E)$ — suppose

$\mathsf{T}(P)^c \subseteq \mathsf{T}(P \setminus_{\mathsf{E}} E)^c \cup B$

$\forall \beta \in B.\ \beta \in \mathsf{V}(H) \cup \{\alpha\}$ — def. E

$\Rightarrow \forall \beta \in B.\ \beta \in \mathsf{V}(H)$ $\beta \neq \alpha$

$\Rightarrow \mathsf{T}(P \setminus_{\mathsf{E}} E)^c \cup B \subseteq \mathsf{V}(H) \cup \{\alpha\}$

$\Rightarrow \mathsf{T}(P)^c \subseteq \mathsf{V}(H)$ (8)

(2.III), $\Rightarrow$

$\mathsf{V}(H) = \mathsf{S}(P)$ suppose

want to show: $\mathsf{V}(H \cup E) = \mathsf{S}(P \setminus_{\mathsf{E}} E)$

$\mathsf{V}(H \cup E) = \mathsf{V}(H) \cup \{\alpha\} = \mathsf{S}(P) \cup \{\alpha\}$

$\mathsf{S}(P \setminus_{\mathsf{E}} E) = \mathsf{S}(P) \cup \{\alpha\}$ def. E

$\Rightarrow \mathsf{S}(P \setminus_{\mathsf{E}} E) = \mathsf{V}(H \cup E)$

(2.III), $\Leftarrow$

$\mathsf{V}(H \cup E) = \mathsf{S}(P \setminus_{\mathsf{E}} E)$ suppose

$\mathsf{V}(H) = \mathsf{V}(H \cup E) \setminus \{\alpha\}$

$\mathsf{S}(P \setminus_{\mathsf{E}} E) = \mathsf{S}(P) \cup \{\alpha\}$ def. E

$\Rightarrow \mathsf{S}(P \setminus_{\mathsf{E}} E) \setminus \{\alpha\} = \mathsf{S}(P)$

$\Rightarrow \mathsf{V}(H) = \mathsf{S}(P)$

(2.IV), $\Rightarrow$

$\mathsf{V}(P) = \mathsf{S}(P) \cup \mathsf{T}(P)$ suppose (9)

want to show: $\mathsf{V}(P) = \mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \mathsf{T}(P \setminus_{\mathsf{E}} E)$

$\mathsf{S}(P) \subseteq \mathsf{S}(P \setminus_{\mathsf{E}} E) \wedge \mathsf{T}(P) \subseteq \mathsf{T}(P \setminus_{\mathsf{E}} E)$ straightforward

Remains to show: $\mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \mathsf{T}(P \setminus_{\mathsf{E}} E) \subseteq \mathsf{S}(P) \cup \mathsf{T}(P)$

$x \in \mathsf{V}(P \setminus_{\mathsf{E}} E)$ suppose

Subcase $x \in \mathsf{S}(P \setminus_{\mathsf{E}} E)$.

$x \in \mathsf{V}(P)$

$\Rightarrow x \in \mathsf{S}(P) \cup \mathsf{T}(P)$ 9 (10)

Subcase $x \in \mathsf{T}(P \setminus_{\mathsf{E}} E)$.

$x \in \mathsf{V}(P)$

$\Rightarrow x \in \mathsf{S}(P) \cup \mathsf{T}(P)$ 9

(2.IV), $\Leftarrow$

$\mathsf{V}(P) = \mathsf{V}(P \setminus_{\mathsf{E}} E) = \mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \mathsf{T}(P \setminus_{\mathsf{E}} E)$ suppose

want to show: $\mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \mathsf{T}(P \setminus_{\mathsf{E}} E) = \mathsf{S}(P) \cup \mathsf{T}(P)$

$\mathsf{S}(P) \cup \mathsf{T}(P) \subseteq \mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \mathsf{T}(P \setminus_{\mathsf{E}} E)$

remains to show: $\mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \mathsf{T}(P \setminus_{\mathsf{E}} E) \subseteq \mathsf{S}(P) \cup \mathsf{T}(P)$

$x \in \mathsf{V}(P)$ suppose

Subcase $x \in \mathsf{S}(P \setminus_{\mathsf{E}} E)$.

$\mathsf{S}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P) = \{\alpha\}$ def. E

Subcase $x = \alpha$.

$\alpha \notin \mathsf{T}(P)$ assume

$\alpha \in \mathsf{V}(H)$ partial slice (II)

$(\beta, \alpha) \in \mathsf{E}(P)$ def. E

$\Rightarrow \bot$ contra partial slice (V) (11)

$\Rightarrow \alpha \in \mathsf{T}(P)$

$\Rightarrow x \in \mathsf{S}(P) \cup \mathsf{T}(P)$

Subcase $x \in \mathsf{S}(P)$.

$\Rightarrow x \in \mathsf{S}(P) \cup \mathsf{T}(P)$

Subcase $x \in \mathsf{T}(P \setminus_{\mathsf{E}} E)$.

$\mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{T}(P) \subseteq B$ def. E

Subcase $x \in B$.

$B \subseteq \mathsf{V}(H \cup E)$ def. E (12)

$\alpha \notin B$

$\Rightarrow B \in \mathsf{S}(P)$

$\Rightarrow x \in \mathsf{S}(P) \subseteq \mathsf{S}(P) \cup \mathsf{T}(P)$

Subcase $x \in \mathsf{T}(P)$.

$\Rightarrow x \in \mathsf{S}(P) \cup \mathsf{T}(P)$

$\Rightarrow \mathsf{T}(P \setminus_{\mathsf{E}} E) \subseteq \mathsf{S}(P) \cup \mathsf{T}(P)$ (13)

(2.V), $\Rightarrow$

H, P, G a partial slice suppose

$\forall \beta \in \mathsf{V}(G).\ \nexists(\beta, \alpha) \in \mathsf{E}(G)$ $\alpha \in \mathsf{S}(G)$ (14)

$\forall (x, y) \in \mathsf{E}(G).y \notin \mathsf{V}(H \cup E)$ def. partial slice, 14

$\mathsf{V}(H \cup E) = \mathsf{V}(H) \cup \{\alpha\}$ def. E

$(x, y) \in \mathsf{E}(P)$ suppose

Subcase $(x, y) \in \mathsf{E}(P) \setminus_{\mathsf{E}} E$.

$y \notin \mathsf{V}(H)$

$y \neq \alpha$ $(x, y) \notin E$

$\Rightarrow y \notin \mathsf{V}(H \cup E)$

Subcase $(x, y) \in E$.

$\Rightarrow (x, y) \notin \mathsf{E}(P) \setminus_{\mathsf{E}} E$

(2.V), $\Leftarrow$

$H \cup E, P \setminus_{\mathsf{E}} E, G$ a partial slice suppose

$\forall (\alpha, \beta) \in \mathsf{E}(G) \cup \mathsf{E}(P \setminus_{\mathsf{E}} E).\ \beta \notin \mathsf{V}(H \cup E)$ def. partial slice

$\forall (\alpha, \beta) \in \mathsf{E}(G) \cup \mathsf{E}(P \setminus_{\mathsf{E}} E).\ \beta \notin \mathsf{V}(H)$ immediate

$(\beta, \alpha) \in E \subset \mathsf{E}(P \setminus_{\mathsf{E}} E) \cup E = \mathsf{E}(P)$ suppose

$\alpha \notin \mathsf{V}(H)$ def. E

$\Rightarrow \forall (\alpha', \beta') \in \mathsf{E}(P)$

$\Rightarrow \forall (\alpha', \beta') \in \mathsf{E}(G) \cup \mathsf{E}(P).\ \beta \notin \mathsf{V}(H)$

(2.VI)

Suffices to show: $\mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \{\alpha\} = \mathsf{T}(P) \setminus \mathsf{S}(P)$

$E = \mathsf{in}_P(\alpha) \times \{\alpha\}$

Subcase $\mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \{\alpha\} \subseteq \mathsf{T}(P) \setminus \mathsf{S}(P)$.

Subcase $x = \alpha$.

$\alpha \notin \mathsf{S}(P)$ $E \neq \varnothing$

$\alpha \in \mathsf{T}(P)$ P has depth 1

$\Rightarrow x \in \mathsf{T}(P) \setminus \mathsf{S}(P)$

Subcase $x \neq \alpha \wedge x \in \mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P \setminus_{\mathsf{E}} E)$.
$\mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{T}(P) \subseteq \mathsf{in}_P(\alpha)$
$\mathsf{in}_P(\alpha) \subseteq \mathsf{S}(P) \subseteq \mathsf{S}(P \setminus_{\mathsf{E}} E)$
$\Rightarrow x \in \mathsf{T}(P)$ $x\mathsf{T}(P \setminus_{\mathsf{E}} E) \wedge \in \notin \mathsf{in}_P(\alpha)$
$\mathsf{S}(P \setminus_{\mathsf{E}} E) \supseteq \mathsf{S}(P)$
$x \notin \mathsf{S}(P)$ $x \notin \mathsf{S}(P \setminus_{\mathsf{E}} E)$
$\Rightarrow x \in \mathsf{T}(P) \setminus \mathsf{S}(P)$
$\Rightarrow \mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \{\alpha\} \subseteq \mathsf{T}(P) \setminus \mathsf{S}(P)$
Subcase $\mathsf{T}(P) \setminus \mathsf{S}(P) \subseteq \mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \{\alpha\}$.
$x \in \mathsf{T}(P) \setminus \mathsf{S}(P)$ suppose
$x \notin \mathsf{S}(P)$ (15)
$x \in \mathsf{T}(P) \subseteq \mathsf{T}(P \setminus_{\mathsf{E}} E)$
Subcase $x \notin \mathsf{S}(P \setminus_{\mathsf{E}} E)$.
$\Rightarrow x \in \mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P \setminus_{\mathsf{E}} E) \subseteq \mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \{\alpha\}$
Subcase $x \in \mathsf{S}(P \setminus_{\mathsf{E}} E)$.
$x \in \mathsf{S}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P)$
$\mathsf{S}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P) = \{\alpha\}$
$\Rightarrow x = \alpha \in \mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \{\alpha\}$
$\Rightarrow \mathsf{T}(P) \setminus \mathsf{S}(P) \subseteq \mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \{\alpha\}$
$\Rightarrow \mathsf{T}(P) \setminus \mathsf{S}(P) = \mathsf{T}(P \setminus_{\mathsf{E}} E) \setminus \mathsf{S}(P \setminus_{\mathsf{E}} E) \cup \{\alpha\}$

□

Lemma A.13. *Suppose H, P, G a partial slice of G_0, then:*
(1) $\blacktriangle^*_{G \cup P \cup H}(\mathsf{S}(H)) = \blacktriangle^*_{(G \setminus_{\mathsf{E}} E) \cup (P \cup E) \cup H}(\mathsf{S}(H))$
(2) $\blacktriangle^*_{G \cup P \cup H}(\mathsf{S}(H)) = \blacktriangle^*_{G \cup (P \setminus_{\mathsf{E}} E) \cup (H \cup E)}(\mathsf{S}(H \cup E))$

Proof. (1) $G_0 = G \cup P \cup H = (G \setminus_{\mathsf{E}} E) \cup (P \cup E) \cup H$
(2) $G_0 = G \cup P \cup H = G \cup (P \setminus_{\mathsf{E}} E) \cup (H \cup E)$, and $\mathsf{S}(H) = \mathsf{S}(H \cup E)$

□

Proposition A.14. *Suppose H, P, G a partial slice of G_0:*

$$(\exists H'.\ H, P, G\ \mathsf{suffE}\ H'\ \textit{with}\ \mathsf{V}(H') = Y) \iff \blacktriangle^*_{G_0}(\mathsf{S}(H)) = Y$$

Proof.
H, P, G a partial slice suppose (16)
$\Rightarrow$ **direction:**
$H, P, G\ \mathsf{suffE}\ H'$ with $\mathsf{V}(H') = Y$ suppose, $\exists H'$
By induction on derivation of suffE.

Subcase $\dfrac{\text{done}}{H, P, G\ \mathsf{suffE}\ H}\ \mathsf{S}(G) \cap \mathsf{V}(P) \subseteq \mathsf{S}(P) \wedge \mathsf{V}(H) \subseteq \mathsf{T}(G)$

$\Rightarrow \blacktriangle^*_{G_0}(\mathsf{S}(H)) = \mathsf{V}(H)$ Lemma A.11

Subcase $\dfrac{H, P \cup E, G \setminus_{\mathsf{E}} E \ \mathsf{suffE}\ H'}{H, P, G \ \mathsf{suffE}\ H'}$ (pending) $\alpha \in \mathsf{V}(H) \wedge E = \mathsf{outE}_G(\alpha) \neq \varnothing$

$H, P \cup E, G \setminus_{\mathsf{E}} E$ a partial slice of G_0 — Lemma A.12

$\blacktriangle^*_{(G \setminus_{\mathsf{E}} E) \cup (P \cup E) \cup H}(\mathsf{S}(H)) = Y$ — IH

$\Rightarrow \blacktriangle^*_{G_0}(\mathsf{S}(H)) = Y$ — Lemma A.13(i)

Subcase $\dfrac{H \cup E, P \setminus_{\mathsf{E}} E, G \ \mathsf{suffE}\ H'}{H, P, G \ \mathsf{suffE}\ H'}$ (extend) $\alpha \in \mathsf{S}(G) \wedge E = \mathsf{inE}_P(\alpha) \neq \varnothing$

$H \cup E, P \setminus_{\mathsf{E}} E, G$ a partial slice of G_0 — Lemma A.12

$\blacktriangle^*_{G \cup (P \setminus_{\mathsf{E}} E) \cup (H \cup E)}(\mathsf{S}(H \cup E))$ — IH

$\Rightarrow \blacktriangle^*_{G_0}(\mathsf{S}(H))$ — Lemma A.13(ii)

$\Leftarrow$ direction

By strong lexicographical induction over $(|\mathsf{E}(G)|, |\mathsf{E}(P)|)$

$\blacktriangle^*_{G_0}(\mathsf{S}(H)) = Y$ — suppose

Subcase $\mathsf{S}(G) \cap \mathsf{V}(P) \subseteq \mathsf{S}(P) \wedge \mathsf{V}(H) \subseteq \mathsf{T}(G)$.

$\dfrac{}{H, P, G \ \mathsf{suffE}\ H}$ (done) $\mathsf{S}(G) \cap \mathsf{V}(P) \subseteq \mathsf{S}(P) \wedge \mathsf{V}(H) \subseteq \mathsf{T}(G)$ — def. suffE

$\Rightarrow \blacktriangle^*_{G_0}(\mathsf{S}(H)) = \mathsf{V}(H) = Y$ — Lemma A.11

Subcase $\alpha \in \mathsf{V}(H) \wedge E = \mathsf{outE}_G(\alpha) \neq \varnothing$. — $\exists \alpha, \exists E$

$(|\mathsf{E}(G \setminus_{\mathsf{E}} E)|, |\mathsf{E}(P)|) < (|\mathsf{E}(G)|, |\mathsf{E}(P \cup E)|)$

$H, P \cup E, G \setminus_{\mathsf{E}} E$ a partial slice of G_0 — Lemma A.12

$H, P \cup E, G \setminus_{\mathsf{E}} E \ \mathsf{suffE}\ H'$ — IH; $\exists H'$

$\dfrac{H, P \cup E, G \setminus_{\mathsf{E}} E \ \mathsf{suffE}\ H'}{H, P, G \ \mathsf{suffE}\ H'}$ (pending) $\alpha \in \mathsf{V}(H) \wedge E = \mathsf{outE}_G(\alpha) \neq \varnothing$ — def. suffE

$\blacktriangle^*_{(G \setminus_{\mathsf{E}} E) \cup (P \cup E) \cup H}(\mathsf{S}(H)) = \blacktriangle^*_{G \cup P \cup H}(\mathsf{S}(H))$ — Lemma A.13(i)

$\Rightarrow \blacktriangle^*_{G_0}(\mathsf{S}(H)) = \mathsf{V}(H') = Y$ — IH

Subcase $\alpha \in \mathsf{S}(G) \wedge E = \mathsf{inE}_P(\alpha) \neq \varnothing$. — $\exists \alpha, \exists E$

$(|\mathsf{E}(G)|, |\mathsf{E}(P \setminus_{\mathsf{E}} E)|) < (|\mathsf{E}(G)|, |\mathsf{E}(P)|)$

$H \cup E, P \setminus_{\mathsf{E}} E, G$ a partial slice of G_0 — Lemma A.12

$H \cup E, P \setminus_{\mathsf{E}} E, G \ \mathsf{suffE}\ H'$ — IH, $\exists H'$

$\dfrac{H \cup E, P \setminus_{\mathsf{E}} E, G \ \mathsf{suffE}\ H'}{H, P, G \ \mathsf{suffE}\ H'}$ (extend) $\alpha \in \mathsf{S}(G) \wedge E = \mathsf{inE}_P(\alpha) \neq \varnothing$ — def. suffE

$\blacktriangle^*_{G \cup P \cup H}(\mathsf{S}(H)) = \blacktriangle^*_{G \cup (P \setminus_{\mathsf{E}} E) \cup (H \cup E)}(\mathsf{S}(H \cup E))$ — Lemma A.13(ii) (17)

$\Rightarrow \blacktriangle^*_{G_0}(\mathsf{S}(H)) = \mathsf{V}(H') = Y$ — IH

□

B SOURCE CODE FOR DATA VISUALISATION FIGURES AND BENCHMARKED PROGRAMS

```
let isCountry name x = name == x.country;
    isYear year x = year == x.year;

let plot year countries =
  let rens = filter (isYear year) renewables;
      nonRens = filter (isYear year) nonRenewables;
  let plotCountry country =
    let rens' = filter (isCountry country) rens;
        rensOut = sum (map (fun x → x.output) rens');
        rensCap = sum (map (fun x → x.capacity) rens');
        x = head (filter (isCountry country) nonRens);
        nonRensCap = x.nuclearCap + x.petrolCap + x.gasCap + x.coalCap
    in {
      x: rensCap / (rensCap + nonRensCap),
      y: (rensOut + x.nuclearOut) / (rensCap + x.nuclearCap)
    }
  in map plotCountry countries

in ScatterPlot {
  caption: "Clean energy efficiency vs. proportion of renewable energy capacity",
  data: plot 2018 [ "BRA", "CHN", "DEU", "FRA", "EGY", "IND", "JPN", "MEX", "NGA", "USA" ],
  xlabel: "Renewables/TotalEnergyCap",
  ylabel: "Clean Capacity Factor"
}
```

Fig. 14. Source code: `energy-query` (Figure 2 that visualises linked inputs)

```
MultiPlot {
   "stacked-bar-chart" :=
      let totalFor year country rows =
         let [ row ] = [ row | row ⟵ rows, row.year == year, row.country == country ]
         in row.nuclearOut + row.gasOut + row.coalOut + row.petrolOut;
      let stack year = [ { y: country, z: totalFor year country nonRenewables }
                       | country ⟵ ["BRA", "EGY", "IND", "JPN"] ]
      in BarChart {
         caption: "Non-renewables by country",
         data: [ { x: numToStr year, bars: stack year }
               | year ⟵ [2014..2018] ]
      },
   "scatter-plot" :=
      let isCountry name x = name == x.country;
         isYear year x = year == x.year;

      let plot year countries =
      let rens = filter (isYear year) renewables;
            nonRens = filter (isYear year) nonRenewables;
      let plotCountry country =
         let rens' = filter (isCountry country) rens;
            rensOut = sum (map (fun x ⟶ x.output) rens');
            rensCap = sum (map (fun x ⟶ x.capacity) rens');
            x = head (filter (isCountry country) nonRens);
            nonRensCap = x.nuclearCap + x.petrolCap + x.gasCap + x.coalCap
         in {
            x: rensCap / (rensCap + nonRensCap),
            y: (rensOut + x.nuclearOut) / (rensCap + x.nuclearCap)
         }
      in map plotCountry countries

      in ScatterPlot {
         caption: "Clean energy efficiency vs proportion of renewable energy capacity",
         data: plot 2018 [ "BRA", "CHN", "DEU", "FRA", "EGY"
                         , "IND", "JPN", "MEX", "NGA", "USA" ],
         xlabel: "Renewables/TotalEnergyCap",
         ylabel: "Clean Capacity Factor"
      }
}
```

Fig. 15. Source code: `stacked-bar-scatter-plot` (Figure 3 that visualises linked outputs)

```
let filter =
   let edgeDetect = [[0,  1, 0],
                     [1, -4, 1],
                     [0,  1, 0]] in
   ⟦ nth2 i j edgeDetect | (i, j) in (3, 3) ⟧;
convolve input_image filter extend
```

(a) edge-detect

```
let filter =
   let emboss = [[-2, -1, 0],
                 [-1,  1, 1],
                 [ 0,  1, 2]] in
   ⟦ nth2 i j emboss | (i, j) in (3, 3) ⟧;
convolve input_image filter zero
```

(b) emboss

```
let filter =
   let gaussian = [[1,  4, 1],
                   [4, 16, 4],
                   [1,  4, 1]] in
   ⟦ nth2 i j gaussian | (i, j) in (3, 3) ⟧;
convolve input_image filter zero
```

(c) gaussian

```
let zero n = const n;
    extend n = min (max n 1);
let convolve image kernel method =
    let ((m, n), (i, j)) = (dims image, dims kernel);
        (half_i, half_j) = (i `quot` 2, j `quot` 2);
        area = i * j
    in  ⟦ let weightedSum = sum [
            image!(x, y) * kernel!(i' + 1, j' + 1)
            | (i', j') ← range (0, 0) (i - 1, j - 1),
              let x = method (m' + i' - half_i) m,
              let y = method (n' + j' - half_j) n,
              x >= 1, x <= m, y >= 1, y <= n
         ] in weightedSum `quot` area
         | (m', n') in (m, n) ⟧;
let input_image =
   let image = [[15, 13,  6, 9, 16],
                [12,  5, 15, 4, 13],
                [14,  9, 20, 8,  1],
                [ 4, 10,  3, 7, 19],
                [ 3, 11, 15, 2,  9]] in
   ⟦ nth2 i j image | (i, j) in (5, 5) ⟧;
```

(d) Library code

Fig. 16. Source code: convolution examples

```
let year = 2015;
let exclude countries yearData =
   flip map yearData
            (second (filter (fun (country, countryData) → not (elem country countries))))
in caption ("Renewables (GW) by country and energy type, " ++ numToStr year)
   (groupedBarChart True colours1 0.2
      (exclude [] (lookup year data)))
```

(a) `grouped-bar-chart`

```
let year1 = head data;
let addTotal kvs =
   ("Total", sum (map snd kvs)) : kvs;
let countryData country =
   map (second (compose addTotal (lookup country))) data in
(fun country →
   caption country
      (lineChart True ("black" : colours1) (fst year1)
         (countryData country)))
"China"
```

(b) `line-chart`

```
let year = 2015 in
caption ("Total renewables (GW) by country, " ++ numToStr year)
   (stackedBarChart True colours1 0.2 (lookup year data))
```

(c) `stacked-bar-chart`

Fig. 17. Source code: graphics examples

```
let coords (Group gs) = Point 0 0;
    coords (Rect x y _ _ _) = Point x y;
    coords (Text x y _ _ _) = Point x y;
    coords (Viewport x y _ _ _ _ _ _ _) = Point x y;
let get_x g = let Point x _ = coords g in x;
let get_y g = let Point _ y = coords g in x;
let set_x x (Group gs) = error "Group has immutable coordinates";
    set_x x (Rect _ y w h fill) = Rect x y w h fill;
    set_x x (Text _ y str anchor baseline) = Text x y str anchor baseline;
    set_x x (Viewport _ y w h fill margin scale translate g) =
                Viewport x y w h fill margin scale translate g;
let dimensions2 (Point x1 y1, Point x2 y2) = Point (max x1 x2) (max y1 y2);
let dimensions (Group gs) = foldl (curry dimensions2) (Point 0 0) (map (coords_op) gs);
    dimensions (Polyline ps _ _) = foldl (curry dimensions2) (Point 0 0) ps;
    dimensions (Rect _ _ w h _) = Point w h;
    dimensions (Text _ _ _ _ _) = Point 0 0;
    dimensions (Viewport _ _ w h _ _ _ _ _) = Point w h;
let coords_op g =
        let (Point x y, Point w h) = prod coords dimensions g in
        Point (x + w) (y + h);
let width g = let Point w _ = dimensions g in w;
let height g = let Point _ h = dimensions g in h;
let spaceRight z sep gs =
   zipWith set_x (iterate (length gs) ((+) sep) z) gs;
let colours1 = [ "#66c2a5", "#a6d854", "#ffd92f", "#e5c494"
               , "#fc8d62", "#b3b3b3", "#8da0cb", "#e78ac3"];
let colours2 = [ "#e41a1c", "#377eb8", "#4daf4a", "#984ea3"
               , "#ff7f00", "#ffff33", "#a65628", "#f781bf"];
let scaleToWidth w (Rect x y _ h fill) = Rect x y w h fill;
    scaleToWidth w (Viewport x y w0 h fill margin (Scale x_scale y_scale) translate g) =
      let scale = Scale (x_scale * w / w0) y_scale in
      Viewport x y w h fill margin scale translate g;
let stackRight sep gs =
   map (scaleToWidth (1 - sep)) (spaceRight (sep / 2) 1 gs);
let groupRight sep gs =
   Viewport 0 0 (length gs) (maximum (map height gs)) "none" 0 (Scale 1 1)
           (Translate 0 0) (Group (stackRight sep gs));
let tickEvery n =
   let m = floor (logBase 10 n) in
   if n <= 2 * 10 ** m then 2 * 10 ** (m - 1) else 10 ** m;
```

Fig. 17. Source code: library code for graphics examples (1 of 3)

```
let axisStrokeWidth = 0.5;
    axisColour = "black";
    backgroundColour = "white";
    defaultMargin = 24;
    markerRadius = 3.5;
    tickLength = 4;
let tick Horiz colour len = Line (Point 0 0) (Point 0 (0 - len)) colour axisStrokeWidth;
    tick Vert colour len = Line (Point 0 0) (Point (0 - len) 0) colour axisStrokeWidth;
let label Horiz x distance str = Text x (0 - distance - 4) str "middle" "hanging";
    label Vert x distance str = Text (0 - distance) x str "end" "central";
let labelledTick orient colour len str =
   Group [tick orient colour len, label orient 0 len str];
let mkPoint Horiz x y = Point y x;
    mkPoint Vert x y = Point x y;
let axis orient x start end =
   let tickSp = tickEvery (end - start);
       firstTick = ceilingToNearest start tickSp;
       lastTick = ceilingToNearest end tickSp;
       n = floor ((end - firstTick) / tickSp) + 1;
       ys = iterate n ((+) tickSp) firstTick;
       ys = if firstTick > start then start : ys else ys;
       ys = if lastTick > end then concat2 ys [lastTick] else ys;
       ps = map (mkPoint orient x) ys;
       ax = Group [
          Line (head ps) (last ps) axisColour axisStrokeWidth,
          Polymarkers ps (flip map ys (compose (labelledTick orient axisColour tickLength)
                          numToStr))
       ]
   in (ax, lastTick);
let catAxis orient x catValues =
   let ys = iterate (length catValues + 1) ((+) 1) 0;
       ps = map (mkPoint orient x) ys
   in Group [
      Line (head ps) (last ps) axisColour axisStrokeWidth,
      Polymarkers (tail ps) (map (const (tick orient axisColour tickLength)) catValues),
      Polymarkers (flip map (tail ps) (fun (Point x y) ⟶ Point (x - 0.5) y))
                  (map (label orient -0.5 0) catValues)
   ];
```

Fig. 17. Source code: library code for graphics examples (2 of 3)

```
let viewport x_start x_finish y_finish margin gs =
   Viewport 0 0 (x_finish - x_start) y_finish backgroundColour margin
            (Scale 1 1) (Translate (0 - x_start) 0) (Group gs);
let lineChart withAxes colours x_start data =
   let xs = map fst data;
       nCat = length (snd (head data));
   let plot (n, colour) =
      let ps = map (fun (x, kvs) → Point x (snd (nth n kvs))) data
      in Group [
         Polyline ps colour 1,
         Polymarkers ps (repeat (length ps) (Circle 0 0 markerRadius colour))
      ];
   let lines = zipWith (curry plot) (iterate nCat ((+) 1) 0) colours;
       x_finish = last xs;
       y_finish = maximum (flip map data (fun (_, kvs) → maximum (map snd kvs)))
   in if withAxes
      then
         let (x_axis, x_finish) = axis Horiz 0 x_start x_finish;
             (y_axis, y_finish') = axis Vert x_start 0 y_finish
         in viewport x_start x_finish y_finish' defaultMargin (x_axis : y_axis : lines)
      else viewport x_start x_finish y_finish 0 lines;
let categoricalChart plotValue withAxes colours sep data =
   let gs = stackRight sep (plotValue colours (map snd data));
       w = length gs;
       h = maximum (map height gs)
   in if withAxes
      then
         let x_axis = catAxis Horiz 0 (map fst data);
             (y_axis, h') = axis Vert 0 0 h
         in viewport 0 w h' defaultMargin (concat2 gs [x_axis, y_axis])
      else viewport 0 w h 0 gs;
let rects colours ns = zipWith (fun colour n → Rect 0 0 1 n colour) colours ns;
let stackedBar colours ns =
   let heights = map snd ns;
       subtotals = scanl1 (+) 0 heights;
       dims = zip (0 : subtotals) heights;
       rects = map
          (fun ((y, height), colour) → Rect 0 y 1 height colour)
          (zip dims colours)
   in Viewport 0 0 1 (last subtotals) "none" 0 (Scale 1 1) (Translate 0 0) (Group rects);
let barChart = categoricalChart rects;
let groupedBarChart = categoricalChart (compose map (flip (barChart False) 0));
let stackedBarChart = categoricalChart (compose map stackedBar);
let caption str (Viewport x y w h fill margin scale translate g) =
   let g' = Group [ Text (x + w / 2) -2 str "middle" "hanging",
                    Viewport 0 0 w h fill margin scale translate g ]
   in Viewport x y w h backgroundColour (defaultMargin / 2 + 4) (Scale 1 1) (Translate 0 0) g';
```

Fig. 17. Source code: library code for graphics examples (3 of 3)